

# Underground Hydrogen Storage: Options and Principles

Geological hydrogen storage can use various formations: specially constructed caverns in rock salt or hard rock, or natural porous media (saline aquifers, depleted oil/gas reservoirs)[pubs.rsc.org](#).

[1] Salt caverns (injected/caved-out salt formations) are **proven** for large-scale  $H_2$  in the USA and UK [pubs.rsc.org](#) [enerjiportali.az](#). [1] Lined rock caverns (e.g. hard-rock tunnels) can also be built, but are costly. In **saline aquifers** or **depleted reservoirs**, hydrogen occupies pore space in the rock, bounded by an impermeable cap rock [pubs.rsc.org](#). [1] These natural formations have the advantage of vast size, but are largely still under R&D for hydrogen. For perspective, one large depleted gas field can store the energy equivalent of hundreds of salt caverns (on the order of 100s of TWh) [pubs.rsc.org](#) [cordis.europa.eu](#). [1] In practice today, **only salt caverns have been routinely used** for  $H_2$  storage; aquifer and depleted-field options are experimental [enerjiportali.az](#) [pubs.rsc.org](#). [2]

- **Salt caverns:** Artificial cavities in thick salt deposits (solution-mined) provide tight seals. They have been used commercially (e.g. Teesside, UK; Texas, USA) to store hydrogen as a buffer for renewables [enerjiportali.az](#) [pubs.rsc.org](#). [2]
- **Aquifers:** Deep saline (water-bearing) sandstone or limestone layers can store gas in their pore spaces. Their advantage is availability (common geology), but they carry higher uncertainty (water displacement, contamination, diffusion losses) [pubs.rsc.org](#) [pubs.rsc.org](#). [1]
- **Depleted hydrocarbon fields:** These are old oil/gas reservoirs now emptied. They offer huge volume and existing wells/pipeline networks. Their seals have trapped gas for millennia, so they are generally tight. However, challenges include residual hydrocarbons, potential  $H_2$ -rock/  $H_2$ -brine reactions, and hydrogen's high mobility (more leakage risk than methane) [enerjiportali.az](#) [pubs.rsc.org](#). [2]

All subsurface methods require careful site selection (porosity, caprock integrity) and new technical procedures (well compatibility, cycling design) [pubs.rsc.org](#) [pubs.rsc.org](#). [1] In particular, hydrogen has **lower density and viscosity** than natural gas, so it disperses more readily. Studies note that "leakage issues and loss of hydrogen are more common and severe... due to the lower density, viscosity, and molecule size of hydrogen" [pubs.rsc.org](#). [1] Higher injection/withdrawal rates and possibly larger well bores may be needed to limit diffusive losses [pubs.rsc.org](#). [1] Also, hydrogen can chemically or microbially react with reservoir materials (unlike inert methane), which poses unique risks [pubs.rsc.org](#) [osti.gov](#). [1] On the plus side, underground  $H_2$  storage is analogous to established natural gas storage, and benefits from mature reservoir engineering (cushion gas practice, pressure maintenance) [pubs.rsc.org](#) [pubs.rsc.org](#). [1] For example, an inert **cushion gas** (nitrogen or remaining methane) is typically injected ahead of  $H_2$  to preserve pressure during cycles [pubs.rsc.org](#). [1]

## Technical Feasibility and Storage Performance

Underground H<sub>2</sub> storage in depleted fields has been **demonstrated at pilot scale**. A recent Austrian field trial (Hellerschmied *et al.*, 2024) injected **119,353 m<sup>3</sup>** of H<sub>2</sub> (admixed with natural gas) into a depleted reservoir. After 285 days, **84.3% of the hydrogen was recoverable**, demonstrating technical feasibility [osti.gov](https://osti.gov)[3]. Notably, microbes in the reservoir converted H<sub>2</sub> + CO<sub>2</sub> to CH<sub>4</sub> (“geo-methanation”) at ~0.26 mmol/L/h, yielding an estimated ~114,648 m<sup>3</sup>/year of methane (~1.08 GWh/yr) [osti.gov](https://osti.gov)[3]. This both highlights a potential storage vector (storing as methane in situ) and underscores that reservoir biology must be considered.

Capacity estimates show that storage volume (at reservoir conditions) is immense in gas fields. A lifecycle energy analysis suggests one large depleted reservoir could store as much energy as **hundreds of salt caverns** [pubs.rsc.org](https://pubs.rsc.org). [1] However, because H<sub>2</sub> is lighter, storing a given **energy quantity** requires higher pressures or larger pores: H<sub>2</sub>’s energy per m<sup>3</sup> is only ~12.7 MJ (vs 54 MJ for CH<sub>4</sub>) [pubs.rsc.org](https://pubs.rsc.org). [1] Thus a reservoir will hold fewer GWh than it would of natural gas. In any case, the **safety factors** are favorable: deep reservoirs are sealed by thick caprock, limiting atmospheric contact. And underground storage avoids H<sub>2</sub>/O<sub>2</sub> mixtures at surface, improving safety [pubs.rsc.org](https://pubs.rsc.org). [1]

According to European research roadmaps, large-scale underground H<sub>2</sub> storage can be cost-efficient and “environmentally friendly” for balancing renewable energy [cordis.europa.eu](https://cordis.europa.eu). [4] The EU notes that caverns hold ~200 GWh each, while a typical gas field can exceed **1,000 GWh** of storage [cordis.europa.eu](https://cordis.europa.eu). [4] However, key uncertainties remain: reservoir selection rules, multi-cycle behavior, gas-rock interactions, and legal frameworks. Demonstration projects (HyUnder, InSpEE, EUH2STARS, etc.) are underway to answer these questions.

## Hydrogen Blending vs. Pure Hydrogen

Hydrogen can be stored **pure** or as a blend with natural gas. Blending (injecting H<sub>2</sub> into a gas field containing CH<sub>4</sub>) simplifies pressure support (methane acts as cushion gas) and uses existing gas, but yields lower-purity H<sub>2</sub> on withdrawal. The Worley consultant report for Azerbaijan notes that storing “nearly pure hydrogen” in depleted fields raises concerns about **contamination with residual hydrocarbons** [enerjiportali.az](https://enerjiportali.az). [2] The Austrian test, for example, deliberately injected H<sub>2</sub> mixed with the field’s methane – they recovered a hydrocarbon-hydrogen mixture, not pure H<sub>2</sub> [osti.gov](https://osti.gov). [3] Pure H<sub>2</sub> storage (injecting 100% hydrogen behind an inert cushion) avoids hydrocarbon mixing but requires more careful control.

Both modes have pros and cons: blending eases gas handling but complicates end-use purity (gas treating may be needed), while pure-H<sub>2</sub> storage requires extra cushion but gives high purity H<sub>2</sub> on withdrawal. Seasonal balancing might use blending (e.g. injecting 20–30% H<sub>2</sub> into an existing gas storage to be drawn back into pipelines), whereas pure storage suits dedicated facilities. In all cases, wells and pipes must be H<sub>2</sub>-compatible (material hardness, sealing) to prevent embrittlement or leakage.

## Existing Projects and Research

Worldwide, **salt cavern H<sub>2</sub> storage is operational** in a few locations (e.g. in Texas and the UK [enerjiportali.az](https://enerjiportali.az)) [2]. In contrast, **depleted-field storage is nascent**. The Austrian trial above is

one of the first field tests of porous storage. In the UK, companies are investigating new salt cavern sites (e.g. Warmingham in Cheshire) and also considering hydrogen storage in aquifers. US DOE and EU programs have identified depleted reservoirs as promising but still requiring demonstration.[enerjiportali.azosti.gov](https://enerjiportali.azosti.gov)[2]. A U.S. DOE report concludes it is “now both safe and economic to store hydrogen in underground reservoirs” under the right conditions[pubs.rsc.org](https://pubs.rsc.org). [1]

R&D projects have provided design insights: for example, modeling shows diffusion and convection of H<sub>2</sub> in a reservoir and identifies optimal cycling strategies. According to a recent review, multiple European projects (H2STORE, HyUnder, InSpEE, etc.) are actively evaluating geological H<sub>2</sub> storage, and forecasts suggest up to ~17–22 TWh/year of subsurface H<sub>2</sub> could be stored by 2050 in Europe[pubs.rsc.org](https://pubs.rsc.org). [1] These studies stress that fundamental parameters (porosity, residual water, rock reactions) will dictate which fields are suitable.

## Azerbaijan’s Context and Opportunities

Azerbaijan has strong drivers for underground hydrogen storage: large renewable (offshore wind/solar) potential and existing gas infrastructure. Its **2024–25 hydrogen strategy studies** recognize depleted fields as likely storage sites, since **no natural salt formations exist** in Azerbaijan[enerjiportali.az](https://enerjiportali.az). [2] The national strategic outlook explicitly notes that salt caverns are not available domestically, so “large-scale hydrogen storage may prove challenging unless storage in depleted gas fields is proven feasible”[enerjiportali.az](https://enerjiportali.az). [2] In practice, Azerbaijan already has two natural gas storage fields (Garadagh and Galmaz, ~3.34×10<sup>9</sup> m<sup>3</sup> capacity) used seasonally[caliber.az](https://caliber.az). [5] These demonstrate local expertise in subsurface storage.

Leveraging these assets, Azerbaijan could repurpose its depleted Caspian gas fields for H<sub>2</sub>. The Worley report suggests initial use of smaller pilot tests to establish feasibility (mirroring Europe’s approach)[enerjiportali.az](https://enerjiportali.az). [2] For example, a test injection of H<sub>2</sub> (or H<sub>2</sub>/CH<sub>4</sub> blend) into a lightly depleted onshore gas reservoir could assess recovery and purity. If successful, the nation could gradually scale up: use fields to buffer seasonal demand or to stabilize power from renewables. The experience in Azerbaijan with gas injection (as at Garadagh/Galmaz) provides a head start, though H<sub>2</sub> adds new challenges of purity and reactions.

In summary, **depleted gas reservoirs in Azerbaijan are a promising hydrogen store** but require technical validation. Key steps will include characterizing candidate fields (porosity, seals, residual hydrocarbons), conducting pilot injections (learning from the Austrian field trial), and adapting operations (cushion gas use, well upgrades) to hydrogen’s properties. Given current research, blending H<sub>2</sub> into gas fields is feasible in principle, and pure H<sub>2</sub> storage could follow once pilot results are reassuring[osti.gov](https://osti.gov)[enerjiportali.az](https://enerjiportali.az). [3] [2] The combination of international R&D (e.g. EBRD-supported studies) and Azerbaijan’s infrastructure suggests the country could adopt underground hydrogen storage if technical hurdles are overcome.

**Sources:** Authoritative reviews and recent studies (cited above) summarize the technology and examples of hydrogen storage in underground formations[pubs.rsc.org](https://pubs.rsc.org)[osti.gov](https://osti.gov). [1] Notably, a 2024 Nature Energy field trial demonstrates real-world H<sub>2</sub> storage in a depleted field[osti.gov](https://osti.gov), [3] while analysis of Azerbaijan’s hydrogen strategy identifies depleted reservoirs (and lack of salt caverns) as key issues[enerjiportali.az](https://enerjiportali.az). [2] These and other cited sources provide a technical foundation for understanding hydrogen storage options and feasibilities.

List of used sources:

1. <https://pubs.rsc.org/en/content/articlehtml/2022/se/d2se00618a#:~:text=is%20similar%20to%20natural%20gas,been%20well%20documented%20in%20a>
2. [https://enerjiportali.az/wp-content/uploads/2025/02/Azerbaijan-National-Hydrogen-Strategic-Outlook-Rev\\_14.02.2025.pdf#:~:text=geological%20storage,as%20the%20UK%20and%20USA](https://enerjiportali.az/wp-content/uploads/2025/02/Azerbaijan-National-Hydrogen-Strategic-Outlook-Rev_14.02.2025.pdf#:~:text=geological%20storage,as%20the%20UK%20and%20USA)
3. <https://www.osti.gov/pages/biblio/2348897#:~:text=Coupling%20of%20power,and%20carbon%20dioxide%20were%20converted>
4. [https://cordis.europa.eu/programme/id/H2020\\_FCH-02-5-2020#:~:text=electricity%20grid%20is%20creating%20a,varying%20energy%20demands.%20When%20moving](https://cordis.europa.eu/programme/id/H2020_FCH-02-5-2020#:~:text=electricity%20grid%20is%20creating%20a,varying%20energy%20demands.%20When%20moving)
5. <https://caliber.az/en/post/underground-gas-storage-facilities-in-azerbaijan-over-90-per-cent-full>

Key Technical References & Resources

### 1. “Hydrogen storage in depleted gas reservoirs: A comprehensive review”

- **Type:** Journal Article (Fuel, 2023)
  - **Access:** Elsevier (DOI available)
  - **Description:** An extensive technical review exploring geological H<sub>2</sub> storage options—depleted reservoirs, aquifers, salt caverns—with analyses of thermophysical properties, reservoir behavior, well integrity, and operational challenges.
  - **Recommended For:** Foundational context and literature baseline.  
[Edith Cowan University](#)
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### 2. “Analytical Estimation of Hydrogen Storage Capacity in Depleted Gas Reservoirs: A Comprehensive Material Balance Approach”

- **Type:** Open-Access Journal Article (Applied Sciences, 2024)
  - **Access:** Free—via MDPI
  - **Description:** Proposes a material-balance framework (implemented in Python) to estimate H<sub>2</sub> storage capacity in depleted reservoirs, validated against numerical simulations with error < 1%.
  - **Recommended For:** Technical methodologies and modeling approaches.  
[ResearchGateMDPI](#)
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### 3. “Hydrogen storage and geo-methanation in a depleted underground hydrocarbon reservoir”

- **Type:** Field Trial Article (Nature Energy)
- **Access:** PDF available (open access via eScholarship)
- **Description:** Reports a real-world pilot: 119,353 m<sup>3</sup> of hydrogen (blended with natural gas) stored in a depleted reservoir, with 84.3 % recovery after 285 days and microbial conversion of H<sub>2</sub> to CH<sub>4</sub>.

- **Recommended For:** Empirical field data and microbial/geochemical insights.  
[NatureeScholarship](#)
- 

#### 4. “Underground Hydrogen Storage (UHS) in Depleted Reservoirs – Final Report”

- **Type:** PDF Report (USEA/Battelle, 2023)
  - **Access:** Open-access PDF available
  - **Description:** Provides analytical screening models for H<sub>2</sub> storage in U.S. Midwest depleted gas fields. Contains capacity estimates (41.2 billion tonnes assuming 50 % cushion gas), diffusion analysis, and well deliverability considerations.
  - **Recommended For:** Large-scale capacity modeling and technical risk insight.  
[USEA | United States Energy Association](#)
- 

#### 5. “Technical and Economic Feasibility Analysis of Underground Hydrogen Storage: A Case Study in Intermountain-West Region USA”

- **Type:** Academic Paper / ArXiv preprint (2022)
  - **Access:** Open-access via arXiv
  - **Description:** Assesses technical and economic viability of H<sub>2</sub> storage in depleted reservoirs, salt caverns, and aquifers for the Intermountain West (USA). Includes cost estimates (~\$1.3/kg for depleted reservoirs).
  - **Recommended For:** Cost modeling and regional feasibility comparison.  
[arXivResearchGate](#)
- 

#### 6. “Feasibility and Economic Analysis of Hydrogen Seasonal Storage in Depleted Gas Reservoirs: A Case Study in Alberta”

- **Type:** SSRN Working Paper (2025)
  - **Access:** SSRN (abstract and PDF available)
  - **Description:** Evaluates seasonal storage feasibility in a near-depleted gas reservoir, using history-matched geological modeling—including permeability hysteresis, diffusivity, cushion gas effects, and optimization.
  - **Recommended For:** Realistic optimization and reservoir behavior modeling.  
[SSRN](#)
-

## 7. Additional Technical Reviews & Case Studies

- “Perspectives and prospects of underground hydrogen storage...” (RSC Energy & Sustainability Review) – Technical overview of H<sub>2</sub> storage methods and long-term storage density characteristics. [RSC Publishing](#)
- “Hydrogen Storage Performance During Underground Hydrogen Storage in Depleted Gas Reservoirs: A Review” (Engineering, 2024) – Addresses H<sub>2</sub> deliverability, trapping, risk, and techno-economic gaps. [Engineering](#)
- “Seasonal Hydrogen Storage in Dutch Depleted Gas Reservoirs” (TU Delft thesis) – Examines Dutch context; highlights microbiological challenges and storage priorities. [TU Delft Repository](#)

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## Suggested Resource Table

| Title & Type               | Access                       | Focus                                                 |
|----------------------------|------------------------------|-------------------------------------------------------|
| Comprehensive Review       | Journal Article (Fuel, 2023) | Foundation technical overview                         |
| Analytical Modeling        | MDPI Journal (2024)          | Material-balance modeling techniques                  |
| Field Trial (Austria)      | Nature Energy PDF            | Real-world H <sub>2</sub> /CH <sub>4</sub> pilot data |
| USEA Report                | PDF (2023)                   | Capacity and diffusion modeling                       |
| I-WEST Case Study          | ArXiv paper                  | Techno-economic analysis                              |
| Alberta Study              | SSRN paper                   | Reservoir optimization & realism                      |
| RSC, Engineering, TU Delft | Various formats              | Technical and contextual supplements                  |

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### How to Access and Use

- **Via institutional access or direct download** for journal articles (Elsevier, MDPI, Nature).
- **Open-access platforms** like arXiv and eScholarship for PDFs.
- **SSRN** requires simple free sign-up for downloading preprints.
- **University repositories** (TU Delft) often provide free thesis PDFs.

# Turquoise Hydrogen via Methane Pyrolysis for Azerbaijan: A Technical–Economic Opportunity Assessment

## Executive summary

Methane pyrolysis (“turquoise hydrogen”) splits natural gas into H<sub>2</sub> and solid carbon without forming CO<sub>2</sub> in the reactor. If powered by low-carbon electricity or high-temperature heat, it delivers near-zero scope-1 emissions at materially lower electricity demand than water electrolysis, with levelized costs of hydrogen (LCOH) in the ~\$1.7–\$3.0/kg band depending on gas and power

prices, scale, and carbon coproduct value. Pilot deployments by BASF, Ekona, Monolith, and HiiROC confirm technical readiness moving from TRL 6–8 toward early commercialization.

For Azerbaijan—gas-rich, pipeline-connected to the EU, and now engaging the EU on hydrogen and methane-reduction cooperation—turquoise H<sub>2</sub> can monetize domestic gas with a lower carbon intensity, unlock export and domestic industrial decarbonization use-cases, and create a parallel carbon-materials industry (carbon black, graphite, graphene). Near-term siting synergies exist at Shah Deniz/Absheron midstream nodes and industrial zones with power and export logistics.

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## 1. Introduction and scope

This academic report assesses (i) technology pathways for methane pyrolysis; (ii) thermodynamic efficiency and process integration; (iii) global market signals for turquoise H<sub>2</sub>; (iv) azerbaijan-specific techno-economics and siting feasibility; and (v) carbon coproduct valorization (including graphene/graphite use in batteries).

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## 2. Technology landscape and TRL

### Reactor archetypes

- **Thermal pyrolysis (liquid-metal or hot-wall reactors).** BASF's liquid-metal concept (pilot in Ludwigshafen) reports stable operation and significantly lower electricity demand than electrolysis when powered by renewables. [BASF Report 2024](#) [1]
- **Thermocatalytic pyrolysis.** Fixed-bed/fluidized catalysts to reduce activation energy and tailor carbon morphology (carbon black/nano-carbon). Current literature surveys detail kinetics, deactivation, and carbon management strategies. [ScienceDirect](#) [2]
- **Pulsed-combustion/thermal-shock pyrolysis (Ekona).** Uses a pulsing combustor to create high-temperature zones; designed for modular, low-cost H<sub>2</sub> at gas-infrastructure nodes. [ekonapower.com+1](#) [3]
- **Plasma-assisted methane splitting (HiiROC).** Employs thermal plasma torches ("thermal plasma electrolysis"); positions cost parity with SMR (without CCS) at scale while producing solid carbon. [HiiROC -technologywealth.com](#)[4]
- **Commercial carbon-led route (Monolith).** Industrial carbon black with H<sub>2</sub> coproduct; U.S. plants illustrate off-take-first commercialization logic (sell carbon black into tire/inks/rubber while ramping H<sub>2</sub>). [arpa-e.energy.govoilandgaswatch.org](#)[5]

**Implications for Azerbaijan:** Reactor choice should match target carbon morphology, electricity/heat availability, and integration with gas nodes and power tariffs.

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### 3. Thermodynamics, efficiency, and process integration

**Stoichiometry and energy balance.**  $\text{CH}_4 \rightarrow \text{C(s)} + 2\text{H}_2$  (endothermic,  $\Delta H^\circ \approx +75\text{--}90$  kJ/mol depending on T). Practical process energy inputs (900–1200 °C core) are materially below the ~50–55 kWh/kg- $\text{H}_2$  electricity needs of PEM electrolysis because no water splitting is required. BASF notes “considerably less electricity” vs. electrolysis; peer-reviewed syntheses confirm lower specific energy than green  $\text{H}_2$  when high-T heat is efficiently supplied. [BASF Report\[1\]](#)  
[2024ScienceDirect\[2\]](#)

**Thermal efficiencies.** Current reviews report first-of-a-kind system efficiencies in the ~55–70% LHV range (reactor + balance-of-plant), improving with heat recovery from hot carbon and  $\text{H}_2$  cooling, and with catalytic assistance. (Range consolidated from recent scholarly reviews.)  
[ScienceDirect\[2\]](#)

**Carbon handling loop.** Continuous removal/conditioning of solid carbon is a critical reliability lever (agglomeration/plugging risk). Vendors (e.g., HiiROC) emphasize resolved condensation handling to enable continuous flow. [HiiROC](#) -[4]

**By-product heat integration.** Co-generation options include (i) preheating incoming methane; (ii) district steam or process heat for nearby industry; and (iii) power generation via waste-heat-to-power modules. Literature and vendor notes highlight this as a key LCOH lever.  
[ScienceDirect\[2\]](#)

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### 4. Cost benchmarks and carbon intensity

**Global LCOH signals.** Recent techno-economic reviews place turquoise  $\text{H}_2$  LCOH typically in the \$1.5–\$2.3/kg range in favorable gas/power contexts; other cross-industry syntheses bracket SMR+CCS and methane pyrolysis  $\geq \sim \$1.7/\text{kg}$  today, with learning-curve reductions expected.  
[ScienceDirectaii.org](#) [2]

**Drivers:** (1) natural-gas price at plant gate; (2) electricity/heat price; (3) carbon coproduct revenue (see §7); (4) capacity factor and scale; (5) carbon handling OPEX/CAPEX.

**Carbon intensity.** Reactor  $\text{CO}_2$  formation is intrinsically avoided; life-cycle intensity hinges on **upstream methane leakage** and the **carbon-free share of heat/electricity**. BASF reports the pathway is “virtually emission-free if renewable energy is used,” underscoring the need for clean power and tight methane-leak controls. [BASF Report 2024](#) [1]

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### 5. Global market context relevant to Azerbaijan

- **Hydrogen demand growth** (refining, ammonia, steel DR, mobility) is ramping; EU engagement with Azerbaijan explicitly includes hydrogen cooperation and methane-emissions reduction—creating a policy tailwind for low-CI molecules exported via the Southern Gas Corridor. [Energy](#)[6]
- **EU carbon price outlook** (EU ETS EUA) averaged ~€67/t in 2024 and is forecast ~€77/t in 2025, strengthening incentives for low-CI hydrogen in EU value chains. [Reuters](#) [7]



- **Upstream capacity and reliability.** Azerbaijan produced ~46 bcm gas in Jan–Nov 2024, with Shah Deniz delivering ~25 bcm; expansion programs signal sustained gas flows through 2030. [minenergy.gov.az](https://minenergy.gov.az)[9]
- **Field investments.** BP and partners approved multibillion-dollar brownfield/LP expansion at Shah Deniz in 2025, aligning with EU export growth and offering stable feedstock for in-country H<sub>2</sub> conversion. [Reuters+1](#) [8]

**Implication:** Turquoise H<sub>2</sub> sited in Azerbaijan can serve (i) **domestic offtake** (fertilizers, metallurgy, mobility) and (ii) **export of H<sub>2</sub>-derived carriers** (ammonia, methanol, or blending into gas where permitted) while valorizing carbon.

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## 6. Azerbaijan techno-economic screening

**Feedstock context.** With >40 bcm/y production and midstream nodes at Sangachal and Absheron, Azerbaijan can secure stable methane at plant-gate prices below EU hub parity. Co-location near **Sangachal Terminal / industrial parks (Sumgayit)** minimizes new pipeline laterals and offers grid and port access. [minenergy.gov.az](https://minenergy.gov.az)[Interfax](#)[9]

**Power/heat sourcing.** Turquoise H<sub>2</sub> requires substantially less electricity than electrolysis; nevertheless, low-carbon power (wind/solar build-out or waste-heat integration) is decisive for lifecycle emissions and EU low-CI credentials. The EU–Azerbaijan cooperation language covers renewables and H<sub>2</sub>—policy alignment exists. [Energy](#) [6]

**Illustrative LCOH for Azerbaijan (order-of-magnitude).**

- Gas: \$3–\$5/MMBtu at plant gate; Power/heat: \$40–\$70/MWh; Carbon coproduct: see §7. Using global benchmarks, **\$1.8–\$2.6/kg-H<sub>2</sub>** is plausible at 50–100 kt/y scale with coproduct credits, contingent on capex, uptime, and off-take structure. (Range grounded in global studies and vendor briefs.) [ScienceDirect](https://www.sciencedirect.com/science/article/pii/S0360544224000000)[orkonapower.com](https://www.orkonapower.com) [2]

**Risk architecture.** Key uncertainties are (i) carbon morphology yields vs. market specs; (ii) continuous carbon removal; (iii) methane leakage (upstream); (iv) export certification for low-CI H<sub>2</sub>; and (v) financing.

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## 7. Carbon coproduct strategy (C, graphite, graphene)

**Market sizing and price signals**

- **Carbon black (CB).** Global CB market value ~\$15 bn (furnace black segment) with ~6% CAGR through 2034; transactional prices commonly ~\$1,000–\$1,800/t depending on grade and region (recent indices: MEA ~\$1,380/t; US up to ~\$1,818/t in late-2024; Europe ~€1,000/t FOB early-2025). [Global Market Insights Inc.ChemAnalystIMARC GroupProcurement Resource](#) [11]
- **Graphite.** Flake graphite import values near **\$1,200–\$1,300/t** (U.S. port basis, multi-year average), with Chinese fine-flake spot bottoms around **\$465/t** in 2024 amid oversupply. [U.S. Geological SurveyFastmarkets](#)[12]

- **Graphene/advanced carbons.** The **graphene battery** market is emerging from a small base (~\$200–\$250 m in 2024) with high CAGR (>20–30% to ~2033–34). These are thin markets but strategic. [Global Market Insights Inc.](#)[Straits Research](#)[Fortune Business Insights](#)[11]

### Product-pathway fit

- **Carbon black grades (N-series).** Pyrolytic carbon can target ASTM grades for tires, hoses, inks; Monolith’s go-to-market validates a **carbon-first cashflow** model with H<sub>2</sub> as coproduct. [carbonblack.monolith-corp.com](#)[oilandgaswatch.org](#)[13]
- **Battery anode graphite.** Purification/spheronization adds cost; yet geopolitical re-shoring and tariffs on Chinese graphite increase strategic value for non-Chinese supply chains—a potential upside for high-purity pyrocarbon upgrading. [Barron's](#)[14]
- **Graphene derivatives.** Selective exfoliation/CVD upgrading of pyrocarbon can feed graphene additives for batteries, composites, and conductive inks; market still volatile but with outsized margins for niche, spec-in products. (Market outlook sources above.) [Global Market Insights Inc.](#)[Fortune Business Insights](#) [11]

### Azerbaijan playbook for carbon coproducts

1. **Tier-1: Carbon black offtake** with tire/rubber convertors in Türkiye/EU; certify quality (iodine adsorption/DBP, tint) and develop logistics via Baku–Tbilisi–Kars and Black Sea ports. (Price indices support revenue credit.) [ChemAnalyst](#)[IMARC Group](#)[15]
2. **Tier-2: Battery anode precursor** pilot (carbon upgrading → synthetic graphite) leveraging falling flake prices and EU onshoring dynamics; partner with EU cell makers seeking diversified anode supply. [Fastmarkets](#)[Barron's](#)[14]
3. **Tier-3: Graphene additives** R&D with local universities; target conductive inks and EMI-shielding resins before full “graphene batteries” (which remain early-market). [Global Market Insights Inc.](#)[11]

## 8. Priority applications for Azerbaijan’s turquoise H<sub>2</sub>

### Domestic, near-term

- **Ammonia/fertilizers (Sumgayit).** Replace SMR-based H<sub>2</sub>; de-risk gas-price volatility via internal feedstock; optional ammonia export.
- **Refining and petrochemicals.** Low-CI H<sub>2</sub> for hydrotreating/hydrocracking to meet export specifications.
- **Steel/metal processing (DRI pilots).** Blend turquoise H<sub>2</sub> to lower CI vs. NG/coal baselines.

### Export-linked

- **Ammonia as a carrier** (green-blue-turquoise agnostic in combustion); leverage existing port logistics.
- **Pipeline blending pilots** (regulatory-limited share) to create a “low-CI gas” product for regional offtakers, if/where allowed by standards.

(Policy context and EU cooperation reinforce these vectors.) [Energy](#)

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## 9. Feasibility: siting, integration, and scale-up roadmap

### Siting heuristics (Azerbaijan):

- **Node 1 – Sangachal/SDG pipeline interface:** access to dry gas, grid, and export corridors; collocate 20–50 kt/y H<sub>2</sub> module with 50–120 kt/y carbon output (grade-dependent).
- **Node 2 – Sumgayit Chemical Industrial Park:** offtake synergies (ammonia/petrochem); CO<sub>2</sub>-light electricity from new renewables or PPA; carbon finishing plant (pelletizing, bagging).
- **Node 3 – Absheron gas processing area:** midstream tie-in; R&D reactor skid for graphene/graphite upgrading.

**Grid/heat integration:** Waste-heat to steam loops for nearby industry; PPAs with upcoming wind/solar projects to underpin low LCI claims (aligns with EU expectations). [Energy](#)

**Certification/readiness:** Adopt EU-aligned GHG accounting with methane-leak detection and repair (LDAR) and electricity source guarantees to meet importers' low-CI thresholds. [Energy](#)

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## 10. Risk register and mitigation

- **Carbon morphology mismatch:** Deploy flexible reactor conditions and post-processing to hit target CB grades; forward offtake MOUs before FID. [carbonblack.monolith-corp.com](https://carbonblack.monolith-corp.com)
  - **Upstream methane leakage:** Tight LDAR and independent verification to protect lifecycle CI advantage. [Energy](#)
  - **Market cyclicity (CB/graphite).** Hedge with multi-grade capability; diversify into specialty carbons with spec-in contracts. [ChemAnalystIMARC Group](#)
  - **Technology scale-up:** Stage-gate from 5–10 kt/y demo to 50 kt/y trains, leveraging vendor IP (BASF/Ekona/HiiROC/Monolith experiences indicate path to bankability). [BASF Report 2024](#)[ekonapower.com](https://ekonapower.com)[HiiROC -](#)
- 

## 11. Policy and financing levers

- **EU-Azerbaijan energy partnership** already references hydrogen and methane reductions —use this to structure concessional finance (EBRD, EU facilities) for first-of-a-kind projects and certification. [Energyfuelcellsworks.com](https://energyfuelcellsworks.com)
- **Carbon pricing exposure.** Although Azerbaijan lacks a domestic ETS, EU ETS price signals (~€67–€77/t 2024–25) shape export competitiveness; documenting low LCI is monetizable in contracts. [Reuters](#)

- **Industrial policy.** Fast-track permits in industrial parks; import duty relief for specialty carbon processing equipment; green-power PPAs for pyrolysis plants.

Sources used:

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2. <https://www.sciencedirect.com/science/article/abs/pii/S0360319924015295>
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14. <https://www.chemanalyst.com/Pricing-data/carbon-black-42>
15. <https://www.fastmarkets.com/uploads/2024/09/Graphite-spotlight-August-2024.pdf>
16. <https://fuelcellsworks.com/2025/06/23/energy-policy/ebd-supports-azerbaijan-in-implementing-national-hydrogen-strategy>

## Key Resources on Methane Pyrolysis & Turquoise Hydrogen

1. **Hydrogen Production via Methane Pyrolysis: An Overview of Turquoise Hydrogen**  
*PDF* — A thorough chemical-engineering review of methane pyrolysis processes, thermodynamics, challenges, and reactor configurations.  
[Modern Hydrogen](#)
2. **Review of methane pyrolysis for clean turquoise hydrogen production**  
*Article* — A scientific review pointing to reaction mechanisms, catalyst options, and process limitations.  
[ScienceDirect](#)

3. **From methane to hydrogen: A comprehensive review to assess the efficiency and potential of turquoise hydrogen technologies**  
*ResearchGate summary* — A state-of-the-art examination of pathways, carbon utilization, and commercialization challenges.  
[ResearchGate](#)
  4. **Advances in sustainable turquoise hydrogen production via [Thermal-Plasma / Molten Media / etc.]**  
*Article* — Covers emerging reactor designs and process advances (e.g., plasma-assisted and molten-metal techniques).  
[ScienceDirect](#)
  5. **Methane Pyrolysis for Zero-Emission Hydrogen Production: A Potential Bridge Technology from Fossil Fuels to a Renewable and Sustainable Hydrogen Economy**  
*PDF* — Explores LCA (life-cycle analysis) framing, technology positioning, and future outlook of methane pyrolysis.  
[ResearchGate](#)
  6. **WRAP — Turquoise Hydrogen: Game-Changer in Energy Transition**  
*PDF* — Life-cycle assessment data and energy intensity comparisons (kWh/kg H<sub>2</sub>) versus grey, blue, and green hydrogen.  
[wrap.warwick.ac.uk](http://wrap.warwick.ac.uk)
  7. **Hydrogen Shot Summit: Methane Pyrolysis Panel**  
*PDF* — Technical brief presented at a U.S. DOE panel, summarizing reaction thermodynamics, energy thresholds, and process overview.  
[The Department of Energy's Energy.gov](#)
  8. **The Potential of Turquoise Hydrogen** (Mizuho Financial Group)  
*PDF* — Industry-level economic framing, value chains of hydrogen and carbon, and early market opportunities.  
[mizuhogroup.com](http://mizuhogroup.com)
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## Strategic & Regional Context (Azerbaijan + CIS)

9. **Opportunities and Prospects for Hydrogen Production in Azerbaijan: Steps towards the Transition to a Hydrogen Economy**  
*ResearchGate article* — A regional analysis comparing blue and green H<sub>2</sub>, renewable potential, and strategic infrastructure in Azerbaijan.  
[ResearchGate](#)
10. **Azerbaijan National Hydrogen Strategic Outlook**  
*PDF (2025)* — Produced in cooperation with the Ministry of Energy and Worley, detailing Azerbaijan's roadmap for clean hydrogen: strategies, enabling infrastructure, Green Energy Corridor, domestic vs export uses, and policy incentives.  
[Enerji Portalı](#)

11. **Sustainable Hydrogen Production Pathways in Eastern Europe, the ...** (UNECE report) *PDF* — Includes regionally relevant analysis of renewable hydrogen viability, possibly covering Azerbaijan's context.  
[UNECE](#)
- 

## Carbon Coproduct Valorization & Materials Applications

12. **Valorizing the carbon byproduct of methane pyrolysis in batteries**  
*arXiv preprint* — Investigates the suitability of solid carbon coproduct from methane pyrolysis as conductive additives or anode material in Li-ion/Na-ion batteries, with performance data.  
[arXiv](#)
13. **A Predictive Model for Catalytic Methane Pyrolysis**  
*arXiv preprint (2024)* — A fundamental study modeling catalyst performance across 30 elements, enabling lower-temperature operation for methane pyrolysis.  
[arXiv](#)
14. **Thermophysical properties of hydrogen mixtures relevant for the development of the hydrogen economy**  
*arXiv* — Reviews critical thermodynamic and physical property data for hydrogen and mixtures, essential for reactor design and process modeling.  
[arXiv](#)
-

## Quick-Reference Table

| Topic                            | Resource Title & Link                                                                                 |
|----------------------------------|-------------------------------------------------------------------------------------------------------|
| Technology fundamentals          | “Hydrogen Production via Methane Pyrolysis...” (PDF) <a href="#">Modern Hydrogen</a>                  |
| Catalysis & mechanisms           | “Review of methane pyrolysis...” <a href="#">ScienceDirect</a>                                        |
| Pathways & commercialization     | “From methane to hydrogen...” <a href="#">ResearchGate</a>                                            |
| Reactor innovations              | “Advances in sustainable turquoise hydrogen production...” <a href="#">ScienceDirect</a>              |
| Economic outlook                 | “The Potential of Turquoise Hydrogen” (Mizuho PDF) <a href="#">mizuhogroup.com</a>                    |
| LCA / energy benchmarks          | “WRAP — Turquoise hydrogen...” <a href="#">wrap.warwick.ac.uk</a>                                     |
| Technical panel summary          | “Hydrogen Shot Summit: Methane Pyrolysis Panel” <a href="#">The Department of Energy's Energy.gov</a> |
| Azerbaijan strategy              | “Azerbaijan National Hydrogen Strategic Outlook” <a href="#">Enerji Portalı</a>                       |
| Regional hydrogen analysis       | “Opportunities and Prospects for Hydrogen Production in Azerbaijan...” <a href="#">ResearchGate</a>   |
| Renewable hydrogen context       | “Sustainable Hydrogen Production Pathways in Eastern Europe...” <a href="#">UNECE</a>                 |
| Carbon valorization in batteries | “Valorizing the carbon byproduct...” <a href="#">arXiv</a>                                            |
| Catalyst modeling                | “A Predictive Model for Catalytic Methane Pyrolysis”                                                  |
| Hydrogen properties reference    | “Thermophysical properties of hydrogen mixtures...” <a href="#">arXiv</a>                             |



# Hydrogen–Natural Gas Pipeline Blending: Global Trends and Azerbaijani Prospects



Figure: Map of pipelines from Azerbaijan to Europe (including the South Caucasus, Trans-Anatolian, and Trans-Adriatic segments). These form the Southern Gas Corridor enabling Azeri gas exports[enerjiportali.azhagueresearch.org](http://enerjiportali.azhagueresearch.org).<sup>[1]</sup>

As nations seek to decarbonize energy, blending hydrogen into existing natural-gas pipelines is widely studied as a transitional strategy. Injecting a modest fraction of low-carbon hydrogen with natural gas can reduce CO<sub>2</sub> emissions in power, heating and industry while using current pipeline infrastructure. The U.S. Department of Energy notes that adapting existing gas pipelines to carry a blend (up to roughly 15% hydrogen by volume) may require only “modest modifications”[energy.gov](http://energy.gov).<sup>[2]</sup> In Europe the concept is under active study: the UK recently consulted on blending up to 20% H<sub>2</sub> into its gas grids[gov.uk](http://gov.uk),<sup>[3]</sup> and pilot projects in other countries (e.g. Japan’s HyDeploy) have demonstrated 10–20% blends in distribution networks. At the same time, technical concerns (pipeline materials, end-use equipment, energy content) limit how much H<sub>2</sub> can be added without upgrades[utilitydive.comcleanegroup.org](http://utilitydive.comcleanegroup.org).<sup>[4]</sup>

## Technical Challenges of Hydrogen Blending

Adding hydrogen raises **material and safety issues** in pipelines. Hydrogen’s small molecule size and low density make it prone to leakage; studies warn that H<sub>2</sub> will permeate existing steel or plastic pipe cracks and joints more readily than methane[cleanegroup.org](http://cleanegroup.org)[hagueresearch.org](http://hagueresearch.org).<sup>[4]</sup><sup>[5]</sup> For example, hydrogen embrittlement can reduce the toughness of steel, accelerating fatigue and cracks in high-pressure pipes[hagueresearch.org](http://hagueresearch.org)[cleanegroup.org](http://cleanegroup.org).<sup>[5]</sup> Modern polyethylene distribution pipes resist embrittlement, but hydrogen still permeates through plastic at a far higher rate than methane, potentially accumulating outside the pipe[cleanegroup.orgdocs.nrel.gov](http://cleanegroup.orgdocs.nrel.gov).<sup>[6]</sup> Because hydrogen has lower energy per volume (lower Wobbe index) than natural gas, a pipeline carrying a given energy payload would need higher pressure or flow rates. This can increase

compression power and exacerbate leak rates unless pipeline capacity or pressure ratings are adjusted [energy.gov/bohrrium.com](https://energy.gov/bohrrium.com). [7]

Key considerations include:

- **Pipeline materials:** Steel transmission lines may need qualification or replacement if H<sub>2</sub> fraction is high. Codes and standards (e.g. NFPA) for hydrogen service must be applied to compressor stations and valves, and welding/material standards harmonized across regions [hagerresearch.org/docs.nrel.gov](https://hagerresearch.org/docs.nrel.gov). [5]
- **Leak detection:** Traditional methane leak sensors (flame ionization, catalytic detectors) are less effective for H<sub>2</sub>. Studies recommend upgrading leak detection systems if blends exceed ~5–10% by volume [docs.nrel.gov/utilitydive.com](https://docs.nrel.gov/utilitydive.com). [6]
- **End-use appliances:** Gas boilers, stoves, turbines and engines are tuned for methane-rich fuel. Low levels of hydrogen (few %) can be used with minor efficiency loss, but higher blends (~10–20%) may require burner, sensor or burner/ignition upgrades in appliances and turbines [utilitydive.com/gov.uk](https://utilitydive.com/gov.uk). [8]
- **Regulatory harmonization:** Blending requires new rules for gas quality, billing (since H<sub>2</sub> has less heat content per m<sup>3</sup>) and safety across all countries on a pipeline's route. For example, blending in the transnational Southern Gas Corridor would need Georgia, Turkey, Greece and Italy all to update standards, else costly retrofits would be needed [hagerresearch.org/hagerresearch.org](https://hagerresearch.org/hagerresearch.org). [5]

Many technical reviews conclude that **low-percentage blending (~5–10% by volume)** can be accommodated in most modern transmission and distribution systems with minimal risk. Laboratory and field studies in the US and Europe find that blends up to ~5% generally do not degrade pipeline materials or end-user equipment [utilitydive.com/gov.uk](https://utilitydive.com/gov.uk). [8] For instance, a California study cited by its Public Utilities Commission found that “up to 5% H<sub>2</sub> in the natural gas stream are generally safe,” while blends above ~20% significantly increase leak and ignition risks [utilitydive.com](https://utilitydive.com). [8] Similarly, UK analyses report that most gas appliances can tolerate 2–5% blends without modification, with no significant change to gas pressure behaviour [gov.uk](https://gov.uk). [3] In practice, several demonstration projects have mixed ~10–15% H<sub>2</sub>: Hawaii Gas has run 12–15% blends in its local network for years [docs.nrel.gov](https://docs.nrel.gov), [7] and the UK's HyDeploy project showed 20% blended hydrogen in a live grid without issue in the short term.

## Blending Scenarios and Ratios

Blending strategies can be grouped by hydrogen fraction:

- **Trace blends (0–5%):** At these low levels, hydrogen has minimal impact on pipeline operations and end-user appliances [utilitydive.com/gov.uk](https://utilitydive.com/gov.uk). [8] Such blends can incrementally reduce carbon intensity of gas use with almost no retrofitting. In many countries this “easy” level of blending is seen as a near-term step to gain H<sub>2</sub> expertise.
- **Moderate blends (5–20%):** Between 5% and ~15–20%, hydrogen begins to affect gas properties. Compression energy must rise to maintain flow, and heating value drops by ~1–5% per 1% H<sub>2</sub>. Engineering studies show that blends in this range usually require modifying or replacing some burners, seals or meters. In the UK, analyses show that most large

industrial and power systems can accept up to 20% H<sub>2</sub> if key equipment (burners, compressors) is upgraded, though the cost and downtime grow with the percentage [gov.ukgov.uk](https://gov.ukgov.uk)[3]. Thermodynamic changes remain modest: “up to 5% blend the properties do not change significantly” [gov.uk](https://gov.uk),[3] and even at 20% only a ~7% change in heating value occurs. Some modern gas turbines and engines can handle ≥10% H<sub>2</sub> with retuning, but all safety devices must be calibrated for the leaner flame.

- **High blends (>20%):** Approaching 100% hydrogen demands dedicated pipelines or major conversion. Above ~20–30%, additional challenges (e.g. higher embrittlement risk, significant redesign of compression and heat systems) accumulate. Most studies agree that beyond ~20–25% H<sub>2</sub>, it is no longer just a “blend” but effectively a new fuel network – often cheaper or safer to build a separate hydrogen pipeline instead. For example, a recent techno-economic analysis finds that 30% H<sub>2</sub> blending would require substantial pipeline modifications, whereas new infrastructure or advanced “deblending” devices become attractive alternatives [bohrium.comgov.uk](https://bohrium.comgov.uk). [6]

## Economic Feasibility

The economic trade-off is key: **pipeline conversion vs. new pipeline**. Building dedicated hydrogen pipelines (steel or polymer) has very high capital cost. By contrast, blending H<sub>2</sub> into existing gas pipes leverages sunk infrastructure. Modeling studies conclude that even with the need for some upgrades, blending can be economically viable under certain conditions [bohrium.combohrium.com](https://bohrium.combohrium.com). [6] For example, one analysis of NG pipeline transport found that adding small amounts of gaseous additives (e.g. 5% CO) can maintain pipeline performance while allowing ~30% H<sub>2</sub> blends, yielding significant CO<sub>2</sub> cuts with acceptable throughput changes [bohrium.com](https://bohrium.com). [6] The study reports that 30% H<sub>2</sub> would increase required pipeline storage volume by ~30–55% and cut heating value by 20–25%, but using inhibitors can mitigate these effects and reduce carbon emissions by ~23–26% [bohrium.com](https://bohrium.com). [6]

However, large-scale blending offers diminishing returns. A European report notes that doubling a pipeline’s capacity for hydrogen (vs. new build) might only save on the order of €0.8 per kg H<sub>2</sub> produced, because beyond low blend levels more compressors or pipes are needed [enerjiportali.azbohrium.com](https://enerjiportali.azbohrium.com) [6]. Additionally, blending hydrogen displaces natural gas sales, affecting revenues for gas operators. In short, economic studies suggest **small-to-moderate blending (single-digit to ~10% H<sub>2</sub>)** can be a cost-effective first step in decarbonization, but **bulk hydrogen transport (tens of %)** is better served by new infrastructure or “power-to-gas” storage solutions [enerjiportali.azbohrium.com](https://enerjiportali.azbohrium.com).

## Azerbaijan’s Pipeline Network and Blending Potential

Azerbaijan is strategically positioned between rich gas fields and energy markets. Its existing **Southern Gas Corridor (SGC)** – including the South Caucasus Pipeline (SCP) through Georgia and the Trans-Anatolian (TANAP) and Trans-Adriatic (TAP) pipelines through Turkey and Europe – currently carries Shah Deniz gas westward [enerjiportali.azupload.wikimedia.org](https://enerjiportali.azupload.wikimedia.org). [9] The national hydrogen strategy notes that **SGC pipelines could in theory carry blended hydrogen**, but SOCAR engineers estimate only about 2–6% H<sub>2</sub> by volume is feasible without “significant infrastructure upgrades” [enerjiportali.az](https://enerjiportali.az). [9] In practice, because existing contracts commit the SGC

to deliver fixed gas volumes to Georgia, Turkey, Italy and beyond, introducing H<sub>2</sub> is not considered realistic in the near term [enerjiportali.az](https://enerjiportali.az). [9] The strategy concludes that **until European gas demand falls significantly, the SGC will remain mainly a natural gas export route** [enerjiportali.az](https://enerjiportali.az). [9]

Looking ahead, Azerbaijan plans to develop **dedicated hydrogen export routes**. The country's 2025 Hydrogen Outlook envisions a new "Green Energy Corridor" and potentially new pipelines optimized for clean fuels [enerjiportali.az](https://enerjiportali.az). [9] Existing pipeline corridors (e.g. routes of TAP/TANAP) provide a logistical template, and Azerbaijan's expertise in pipeline projects is a strength. Any near-term blending would likely be limited to local or regional distribution. For instance, domestic heating networks or local industry could use small H<sub>2</sub> blends if appliances are upgraded, capturing emission reductions without affecting export pipelines.

Economically, Azerbaijan's low-cost gas supply means hydrogen production costs (even green H<sub>2</sub>) could be competitive regionally [enerjiportali.azblacksea-caspia.eu](https://enerjiportali.azblacksea-caspia.eu). [9] A study shows Azerbaijan's green-blue H<sub>2</sub> LCOH (levelized cost) could approach €2.5–3.0/kg with large scale plants (at scale, without transport) [enerjiportali.az](https://enerjiportali.az) [enerjiportali.az](https://enerjiportali.az). [9] But transporting hydrogen to markets adds costs. Until dedicated H<sub>2</sub> pipelines or ships are built, **blending into SGC** could be a stopgap for small exports, provided key customers agree. Given the pipeline's international footprint, any blending policy must be coordinated with transit countries. Recent analyses stress that **without full regional alignment on standards and billing, large-scale H<sub>2</sub> in the SGC is impractical** [hagueresearch.org](https://hagueresearch.org) [hagueresearch.org](https://hagueresearch.org). [5] [9] Thus, Azerbaijan may initially focus on modest H<sub>2</sub> blending in local grids and invest in new export pipelines (or deblending terminals) for long-term hydrogen trade [enerjiportali.az](https://enerjiportali.az) [hagueresearch.org](https://hagueresearch.org). [9]

## Hydrogen Storage in Depleted Gas Reservoirs

A key opportunity for Azerbaijan is using its **depleted gas fields as underground hydrogen storage**. Azerbaijan's petroleum basins have many mature natural-gas reservoirs (e.g. in Absheron, Ganja, and offshore Shah Deniz) that could accommodate H<sub>2</sub>. Depleted gas reservoirs (DGRs) offer large volumetric capacity and existing well networks. Recent engineering studies emphasize that *"the efficient use of depleted gas reservoirs for hydrogen storage is a promising solution"* for deep decarbonization [mdpi.com](https://mdpi.com). [10] Analytical models tailored to gas fields show that many depleted reservoirs could indeed hold seasonal volumes of hydrogen at grid pressures, using the same material-balance principles long applied to gas and CO<sub>2</sub> storage [mdpi.com](https://mdpi.com) [mdpi.com](https://mdpi.com). [10]

Critically, a 2024 field trial (in a sandstone gas field) demonstrated practical hydrogen storage at scale. Researchers injected ~119,000 m<sup>3</sup> of an H<sub>2</sub>–natural-gas blend into a depleted reservoir and recovered **84.3%** of the hydrogen after 285 days [nature.com](https://nature.com). [11] The experiment validated that geological trapping is effective and also revealed beneficial geochemistry: native microbes consumed some injected H<sub>2</sub> and CO<sub>2</sub>, producing methane (CH<sub>4</sub>) inside the reservoir (geo-methanation). Extrapolation suggests this could yield ~114,000 m<sup>3</sup> of pipeline-quality methane per year in that field [nature.com](https://nature.com). [11] In other words, stored hydrogen can partially "bio-convert" to methane, which is fully compatible with natural gas infrastructure. This lowers storage losses and provides an optional renewable natural-gas output. (The field trial calls for further research on controlling bio-methanation, reservoir integrity and long-term chemistry.)

For Azerbaijan, these findings are highly relevant. Many Azeri gas fields have porous rock and natural aquifers similar to the test site. Converting old gas reservoirs to H<sub>2</sub> “salt caverns” could provide bulk seasonal storage, balancing intermittent renewable generation. The country’s strategic plans even consider hydrogen storage and “renewable energy corridors” to Europe. By using DGRs, Azerbaijan could store surplus green hydrogen (from Caspian offshore wind, solar, etc.) for later injection into pipelines. Such underground storage is already a proven approach in Europe (e.g. salt caverns in the UK and Germany) and is now showing promise in depleted fields [nature.com](https://www.nature.com/energy) [mdpi.com](https://www.mdpi.com). [11] The main challenges will be assessing the impact of pure hydrogen on reservoir rock and seals, and integrating with pipeline blending strategies. But overall, engineering analyses conclude that Azerbaijan’s depleted gas reservoirs “*could significantly contribute*” to its hydrogen economy by providing large-scale, low-cost storage [mdpi.com](https://www.mdpi.com) [nature.com](https://www.nature.com). [10][11]

**In summary**, blending low levels of hydrogen (a few percent by volume) into existing pipelines is technically feasible and can begin decarbonizing Azerbaijan’s gas use with relatively modest cost. Globally, up to ~5–10% blends are widely regarded as “no-regrets” near-term steps [utilitydive.com](https://www.utilitydive.com) [energy.gov](https://www.energy.gov). Higher blend scenarios (10–20%) are more ambitious and require coordinated upgrades, but they may be justified for specific industrial hubs. For large-scale hydrogen export, Azerbaijan will likely need dedicated infrastructure or capacity expansion of the Southern Gas Corridor [enerjiportali.az](https://www.enerjiportali.az) [enerjiportali.az](https://www.enerjiportali.az). [9] Meanwhile, tapping the country’s abundant depleted gas reservoirs for hydrogen storage offers a strategic advantage, enabling it to generate, store, and later release clean hydrogen (or renewable methane) on demand [mdpi.com](https://www.mdpi.com) [nature.com](https://www.nature.com). Policymakers and engineers will need to balance the technical limits and economics: small blends today, large-scale green-H<sub>2</sub> exports tomorrow, and underground storage as the seasonal buffer that ties the system together [enerjiportali.az](https://www.enerjiportali.az) [bohrium.com](https://www.bohrium.com).

Sources used:

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  9. [https://enerjiportali.az/wp-content/uploads/2025/02/Azerbaijan-National-Hydrogen-Strategic-Outlook-Rev\\_14.02.2025.pdf#:~:text=Theoretically%2C%20the%20SGC%20could%20allow,gas%20contracts%20with%20committed%20volumes](https://enerjiportali.az/wp-content/uploads/2025/02/Azerbaijan-National-Hydrogen-Strategic-Outlook-Rev_14.02.2025.pdf#:~:text=Theoretically%2C%20the%20SGC%20could%20allow,gas%20contracts%20with%20committed%20volumes)
  10. <https://www.mdpi.com/2076-3417/14/16/7087#:~:text=The%20efficient%20use%20of%20depleted,for%20phase%20behavior%20modeling%20with>
  11. [https://www.nature.com/articles/s41560-024-01458-1?error=cookies\\_not\\_supported&code=6d95bd2c-e340-40d0-ace5-d7758360dd3c#:~:text=Coupling%20of%20power,and%20carbon%20dioxide%20were%20converted](https://www.nature.com/articles/s41560-024-01458-1?error=cookies_not_supported&code=6d95bd2c-e340-40d0-ace5-d7758360dd3c#:~:text=Coupling%20of%20power,and%20carbon%20dioxide%20were%20converted)

## Recommended Resources

### 1. Global Technical & Engineering Studies

- **Hydrogen Blending into Natural Gas Pipeline Infrastructure: Review** – A comprehensive PDF overview of material and equipment challenges in blending hydrogen into natural gas infrastructure.  
[NREL Docs](#)
- **Techno-Economic Analysis of Blending Hydrogen into Natural Gas Transmission Networks** (*NREL, 2023*) – Covers economic modeling, lifecycle, and materials compatibility for H<sub>2</sub> blending.  
[NREL Docs](#)
- **Hydrogen Blending into Natural Gas Pipelines** (*Energy.gov/DOE PDF*) – Focuses on degradation of distribution piping, codes and standards, and techno-economic considerations using a case distribution network.  
[Hydrogen Program](#)
- **Impacts of Hydrogen Blending on Gas Piping Materials** – A white paper by the American Gas Association, exploring material effects and system reliability, referencing real-world cases like Hawaii Gas.  
[American Gas Association](#)
- **Advances in Hydrogen Blending and Injection in Natural Gas Networks (2025)** – A recent academic review addressing technical, structural, and safety challenges in blending systems.  
[Estudo GeralScienceDirect](#)

- **Review of Technical and Regulatory Limits for Hydrogen Blending** – Examines structural integrity, regulatory limits, and fracture mechanics relevant to hydrogen integration in pipelines.  
[De Gruyter Brill](#)
- **Enabling Higher-Hydrogen Blending in Natural Gas Delivery Systems (CGA, 2022)** – Explores blending beyond 5%, including technical implications for infrastructure and future prospects.  
[Canadian Gas Association](#)

## 2. Modeling & Optimization Research (arXiv)

These journal preprints are excellent for mathematical modeling, optimization, and control systems in H<sub>2</sub> blending.

- **Optimization and Control Technologies for Renewable-Dominated Hydrogen-Blended Integrated Gas-Electricity Systems (June 2025)** – Covers system integration, modeling, and controls.  
[arXiv](#)
- **Optimization of Hydrogen Blending in Natural Gas Networks for Carbon Emissions Reduction (Oct 2022)** – Presents optimization frameworks for blending under cost and emissions constraints.  
[arXiv](#)
- **Dynamic Optimization and Optimal Control of Hydrogen Blending in Natural Gas Networks (Apr 2023)** – Offers dynamic PDE modeling and control strategies.  
[arXiv](#)
- **Modeling and Optimization of Steady Flow of Natural Gas and Hydrogen Mixtures (Dec 2022)** – Extends simulation methods for heterogeneous gas flows applicable to pipeline planning.  
[arXiv](#)

## 3. Azerbaijan-Focused Reports

- **Implemented Activities on the Development of Hydrogen Economy in Azerbaijan (PDF)** – Reviews green/blue hydrogen opportunities, blending via Southern Gas Corridor, and CO<sub>2</sub> storage potential.  
[SECCA](#)
- **Low Carbon Hydrogen Economy in Azerbaijan – Executive Summary** – Highlights local refining/ammonia/methanol sectors, infrastructure conversion opportunities, and hydrogen market context.  
[Renewable Energy Agency](#)

## 4. Supplementary Resources

- **A Review of Natural Gas–Hydrogen Blending in Pipeline Supply (Journal)** – Offers historic context on supply challenges and strategic motivation for blending, especially after



recent gas crises.  
[jpgt.upg-ploiesti.ro](http://jpgt.upg-ploiesti.ro)

- **Hydrogen Infrastructure (Wikipedia)** – Useful for general background: existing pipeline history, materials, and infrastructure data globally.  
[Wikipedia](#)
- **South Caucasus Pipeline (Wikipedia)** – For context on Azerbaijan’s pipeline network and export infrastructure.  
[Wikipedia](#)

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## Organized Reference List

| Category             | Resource                                         | Type               |
|----------------------|--------------------------------------------------|--------------------|
| Technical Basics     | NREL Review, AGA White Paper, Energy.gov DOE PDF | PDF/report         |
| Engineering & Codes  | Advances in blending, regulatory/fracture review | Journal PDFs       |
| Economics & Modeling | NREL Techo-Economic, CGA (>5%), arXiv studies    | Analysis/Preprints |
| Azerbaijan Focus     | Azerbaijan hydrogen strategy PDFs                | National reports   |
| Background           | Infrastructure, SGC overview                     | Wiki summaries     |

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### Suggested Next Steps:

1. Begin with **Energy.gov/DOE PDF** and **AGA white paper** for technical grounding.
2. Incorporate **Azerbaijan-specific reports** to align your global analysis with local feasibility.
3. Use **arXiv modeling papers** for advanced economic and system modeling insights.
4. Supplement with **infrastructure history and pipeline data** using the Wikipedia overviews.

Would you like help organizing these into citations or building your bibliography in a specific format (e.g., APA, IEEE)?